

TATA, Hungary

WMO: 128360

Lat: 47.650N

Lon: 18.317E

Elev: 129

StdP: 99.78

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.5	-8.7	-14.7	1.1	-9.9	-12.3	1.3	-6.9	10.2	5.1	8.9	3.3	1.1	130	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	33.5	21.7	31.8	20.9	30.1	20.3	22.7	30.9	21.9	29.7	21.0	28.5	2.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	15.0	26.0	19.2	14.2	25.0	18.5	13.5	24.2	67.3	30.9	64.3	29.8	61.3	28.6	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.2	6.2	-14.8	36.2	3.5	1.7	-17.3	37.4	-19.4	38.4	-21.4	39.3	-24.0	40.5		
(5)			-15.0	24.2	3.5	0.9	-17.5	24.8	-19.6	25.3	-21.6	25.8	-24.1	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	-0.2	1.6	6.2	12.1	16.4	20.3	22.0	21.3	16.4	11.3	6.5	1.2		
	DBStd	8.74	4.61	4.66	4.36	3.94	3.73	3.56	3.52	3.32	3.59	4.01	4.29	4.31		
	HDD10.0	1133	316	235	129	23	2	0	0	0	1	34	121	273		
	HDD18.3	2916	574	468	376	189	83	21	8	10	80	220	356	531		
	CDD10.0	1607	0	1	12	86	199	308	371	350	192	73	15	1		
	CDD18.3	348	0	0	0	3	22	79	121	102	21	1	0	0		
	CDH23.3	3758	0	0	0	45	267	830	1311	1075	221	9	0	0		
	CDH26.7	1432	0	0	0	7	64	291	573	444	53	0	0	0		
(14) Wind	WSAvg	2.5	2.5	2.7	3.0	2.6	2.5	2.3	2.3	2.1	2.1	2.3	2.7	2.5		
(15) Precipitation	PrecAvg	584	36	32	34	37	64	61	69	59	55	46	50	41		
	PrecMax	985	77	117	109	86	205	140	165	122	143	109	105	119		
	PrecMin	360	2	0	1	1	14	6	12	1	7	2	4	4		
	PrecStd	131	19	25	24	21	42	32	38	34	36	27	24	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	15.1	21.0	27.5	30.9	33.7	36.5	36.1	30.8	24.6	18.7	13.4		
		MCWB	8.7	9.1	12.5	16.7	18.8	22.0	22.0	21.8	19.7	17.1	13.8	10.0		
	2%	DB	9.7	12.1	18.4	24.4	28.4	31.9	33.9	33.2	28.1	22.2	16.3	10.9		
		MCWB	7.1	7.8	11.0	14.5	18.3	21.1	21.8	21.6	18.6	16.1	12.6	8.5		
	5%	DB	7.7	10.2	15.9	22.4	26.4	30.0	32.1	31.1	26.0	20.3	14.3	8.6		
		MCWB	5.6	7.3	9.8	13.7	17.5	20.4	21.0	20.8	18.0	15.0	11.1	6.7		
	10%	DB	6.1	8.4	13.7	20.3	24.2	28.2	30.1	29.1	23.7	18.2	12.5	7.1		
		MCWB	4.4	5.9	8.7	12.5	16.2	19.6	20.4	20.0	16.7	13.9	9.9	5.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	9.6	13.0	17.2	20.3	23.6	23.6	23.6	20.7	17.9	14.3	10.2		
		MCDB	11.8	14.0	19.1	26.3	28.5	31.8	32.6	31.8	28.6	23.6	17.9	12.8		
	2%	WB	7.3	8.4	11.6	15.4	19.1	22.1	22.7	22.5	19.6	16.6	12.8	8.6		
		MCDB	9.5	11.5	17.1	22.9	26.6	29.6	31.3	30.5	26.4	21.3	16.1	10.7		
	5%	WB	5.8	7.3	10.4	14.1	18.0	21.2	21.9	21.7	18.5	15.5	11.4	7.0		
		MCDB	7.4	9.9	15.2	21.0	24.6	28.3	29.9	29.5	24.5	19.4	13.9	8.5		
	10%	WB	4.5	6.2	9.0	13.1	17.0	20.2	21.0	20.7	17.4	14.4	10.2	5.6		
		MCDB	6.0	8.3	13.2	19.3	23.2	26.7	28.5	27.9	22.4	17.6	12.3	6.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	7.3	9.9	11.9	11.5	11.9	12.5	12.5	11.4	9.8	6.7	5.6		
		MCDBR	8.2	10.1	14.3	15.4	15.2	14.5	15.6	15.7	14.8	13.0	8.7	7.0		
	5% WB	MCWBR	5.9	6.8	8.1	7.2	6.7	5.8	5.7	5.9	6.8	7.3	5.7	5.3		
		MCWBR	7.8	9.2	13.2	14.3	13.6	13.0	13.7	14.1	13.1	11.9	8.1	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.328	0.352	0.389	0.434	0.427	0.428	0.439	0.439	0.414	0.383	0.358	0.327		
		taud	2.363	2.320	2.246	2.138	2.210	2.244	2.215	2.227	2.263	2.328	2.353	2.378		
	Ebn at Noon		738	798	821	816	838	839	822	803	785	750	690	689		
		Edh at Noon	72	93	116	142	137	134	136	129	113	91	73	64		
(44) All-Sky Solar Radiation	RadAvg		1.03	1.84	3.07	4.53	5.45	5.92	5.77	5.17	3.65	2.31	1.21	0.78		
	RadStd		0.12	0.27	0.45	0.44	0.58	0.45	0.44	0.50	0.44	0.25	0.12	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (62 neighbors)	+0.60	N/A	N/A	N/A	+0.54	+0.50	-124	-179	+101	+54

Nomenclature: See separate page